SCANIQ MONTANA LLC

Biomimetic 3D Printing & AI-Driven Innovation for Agricultural Advancement

Grant Application Summary, Talking Points & Strategic Narrative

Prepared for Jacqueline Anderson & John D. Antaramian | July 13, 2026

Tip of the Spear: Livestock Embryonic Lattices, Birthing Technology & Closed-Loop Visual Acuity AI

1. EXECUTIVE SUMMARY (Copy-Paste Ready for Pre-Apps & Narratives)

ScanIQ Montana LLC is establishing a state-of-the-art AI-powered 3D scan-to-print R&D; and manufacturing hub in the Livingston–Lewistown–Yellowstone corridor (within 250 miles / 4-hour radius of Billings) to transform animal husbandry through nature-optimized biomimetic lattice structures, advanced veterinary training models, and AI-integrated breeding support systems. Our patent-pending multi-agentic platform (PicoGK computational geometry + Helix Neural Network for helical/biomimetic toolpath dexterity and visual acuity) converts iPhone LiDAR scans, photogrammetry, or vibe-coded requests into optimized G-code for complex organic geometries that outperform traditional planar slicing in strength-to-weight, material efficiency, and printability on curves/overhangs.

Core Innovation – Visual Acuity & Full Science Team Integration: Cameras mounted at 3D printer nozzle tips plus distributed lab/work area cameras create a real-time closed-loop feedback system. AI agents monitor layer deposition, detect defects (warping, stringing, lattice collapse), auto-adjust parameters or pause for correction, and feed post-print QA data back into Helix/PicoGK for continuous optimization. Lab cameras enable precision livestock phenotyping (body condition scoring, gait/lameness analysis, health metrics via computer vision) that integrates with genotype data for holistic breeding program decisions. This turns passive manufacturing into an active, reasoning member of the science team – accelerating empirical discovery in embryonic development, dystocia management, and advanced reproductive technologies (including cloning support via precision anatomical models and supplemental lattice scaffolds where regulations and ethics permit).

Primary Ag Applications (Tip of the Spear): (1) Realistic bovine/equine birthing simulators with internal embryonic-inspired gyroid/tubular lattice structures mimicking soft tissue and supportive architecture for dystocia training – proven to increase vet/tech/student confidence and procedural success. (2) 3D printed lattice scaffolds for IVF, embryo transfer (ET), and supplemental embryonic development processes that mimic natural extracellular matrix (ECM) for improved viability. (3) Custom animal prosthetics and orthopedic supports: scan injured limb/joint on-ranch → optimized biomimetic lattice design → rapid local print for breeding stock comfort and longevity. (4) AI-phenotype/genotype breeding program infrastructure: camera-driven monitoring + 3D printed custom tools/models to support elite genetic replication and selection. Secondary revenue streams include local manufacturing for ranch tools/parts, custom props for Montana's film industry (Yellowstone productions), and unique biomimetic 3D printed furniture/fixtures for tourism/Airbnb ranch experiences.

Funding Request & Impact: Targeting Montana Growth Through Agriculture (GTA) grants/loans (up to $50k grant / $100k loan, 1:1 match) for equipment (printers with integrated camera systems, multi-axis capability, ag-grade/biocompatible materials), facility setup (lab cameras, climate-controlled print farm), R&D;/pilot validation with local ranches and vets, and personnel for AI breeding integration. Complementary federal opportunities: USDA NIFA Veterinary Services Grant Program (VSGP) Education/Extension/Training (EET) for vet shortage relief via training models and programs; Value-Added Producer Grants (VAPG) for value-adding services/products to livestock producers. Projected outcomes: 20-50% gains in mechanical performance/material efficiency of printed models vs standard; measurable lifts in training competency and embryo viability in pilots; 3-5+ rural high-skill jobs; strengthened local supply chains; positioning Montana as a precision livestock technology leader. All profits fuel the family freedom fund for MS care while building a scalable, patent-protected ag-tech platform with SaaS (Replicator), service, licensing, and hardware upside.

2. PROBLEM STATEMENT & MARKET OPPORTUNITY

Montana's robust cattle and equine industries face persistent, costly challenges that advanced manufacturing and AI can directly mitigate:

• **Dystocia & Reproductive Losses:** Difficult births in cattle and horses result in significant economic losses (vet fees, lost calves/foals, reduced dam productivity, culling). Rural large-animal veterinarian shortages (USDA-designated shortage situations across MT) limit timely intervention and training capacity.

• **Limited Access to Advanced Training Tools:** Existing birthing simulators are expensive, imported, and often lack realistic internal anatomy or biomimetic tissue properties. Vet tech programs and practicing vets need affordable, high-fidelity, locally supported models for dystocia management and advanced repro procedures.

• **Custom Prosthetics & Orthopedics for Breeding Stock:** Elite genetics animals with injuries need rapid, affordable, comfortable custom supports. Current solutions involve long lead times, high costs, and suboptimal fit/performance.

• **Breeding Program Efficiency & Advanced Reproductive Tech (ART):** Improving phenotype selection and embryo viability (IVF/ET, cloning where appropriate) requires better tools, models, and data integration. Rural operations lack access to precision phenotyping and custom fabrication infrastructure.

• **Rural Manufacturing & Supply Chain Gaps:** Ranches, film crews, and tourism operators need fast-turnaround custom parts, props, and fixtures. Importing from distant suppliers is slow and expensive; local advanced mfg capacity creates jobs and resilience.

• **Opportunity:** Montana's film industry (Yellowstone and productions), tourism (Glacier/Yellowstone gateways), and ag sector are hungry for innovative, nature-inspired solutions. A local hub delivering scan-on-ranch → AI-optimized print → deploy can capture high-margin service revenue while generating grant-eligible impact data.

3. SOLUTION: SCANIQ + HELIX NN + VISUAL ACUITY CLOSED-LOOP AI

Our end-to-end pipeline is purpose-built for complex biomimetic and organic geometries that standard slicers (Cura, PrusaSlicer) handle poorly:

Core Pipeline:

• Input: iPhone LiDAR / multi-photo photogrammetry scan of anatomy (pelvis, limb, embryo context) or vibe-coded request ("embryonic lattice birthing model for bovine dystocia training").

• Optimization: PicoGK computational geometry / voxel / mesh processing for clean, analysis-ready models.

• Biomimetic Intelligence: Helix Neural Network (patent-pending extension) applies specialized dexterity (curvature-aware, non-planar/helical toolpaths for lattices, gyroids, trabecular patterns) and acuity (high-res detail preservation, structural optimization) – ideal for embryonic lattices, helical props, bone-inspired supports.

• G-Code Generation: Advanced slicing with minimal supports, scale-aware, material-efficient paths tailored to printer and application.

• **NEW: Visual Acuity Closed-Loop Feedback (Full Science Team Integration):** Nozzle-tip cameras (industrial USB or Pi-based with lighting) provide real-time layer-by-layer monitoring. AI agents detect defects (overhang collapse, stringing in lattice zones, dimensional drift), trigger auto-correction or pause, and log data. Post-print camera QA (or operator-uploaded photos) compares actual vs predicted performance and feeds back to update Helix weights or PicoGK params for future jobs. Lab/work area cameras extend this to live phenotype monitoring of breeding animals – computer vision for body condition, lameness/gait analysis, health indicators – creating a continuous data stream that informs both print optimization and breeding decisions (phenotype + genotype integration). The result: every print and every animal observation makes the entire system smarter. AI becomes a collaborative reasoning partner alongside vets, breeders, and researchers – not a black box.

Why This Wins Grants & Customers: Standard 3D printing in ag/vet is fragmented and planar-limited. We deliver biology-aware, self-improving, closed-loop intelligence at local scale. Empirical targets for pilots: >95% print success rate (vs ~70% standard on complex lattices), 25-40% material savings, 20-50% mechanical performance gains (FEA-validated or physical testing), measurable embryo viability or training competency lifts. The feedback loop provides the rigorous, data-rich narrative funders love.

4. PRIMARY APPLICATIONS (AG TIP OF THE SPEAR) & SECONDARY REVENUE

A. Veterinary Birthing Simulators with Embryonic Lattice Internals

Scan real or stock bovine/equine pelvic anatomy → generate model with internal gyroid/tubular lattice structures that mimic the supportive architecture and tissue compliance of embryonic development environments. Print in durable, realistic materials. Result: High-fidelity training tools that improve dystocia management skills, reduce real-world complications, and boost confidence in vet students, techs, and practitioners. Pilot with local MT vet clinics or extension

programs. Grant alignment: VSGP EET (training/education to relieve vet shortages); GTA (innovative ag tech with scalability and rural impact).

B. Embryonic Development & IVF/ET Support Scaffolds + Cloning Considerations

3D printed lattice scaffolds (embryonic-inspired gyroid, tubular, or trabecular patterns) designed to support in vitro embryo culture, transfer, or supplemental development processes by better mimicking natural ECM architecture – potentially improving viability and implantation rates (backed by existing bioceramic/polymer lattice research in regenerative medicine and embryo tech). For cloning (somatic cell nuclear transfer / advanced ART where commercially practiced in livestock genetics): precision printed anatomical guides, holders, or bioreactor components that aid procedural training or ex vivo support. Frame as "supplemental processes for embryonic development and advanced reproductive technologies" to stay grant-appropriate and ethical. Pilot data with MT partners or MSU can strengthen future claims. This is high-upside IP territory for licensing to genetics companies.

C. Custom Prosthetics, Orthopedics & Precision Ranch Tools

On-ranch scan of injured fetlock, hock, or limb → AI-optimized biomimetic lattice prosthetic/brace (lightweight, strong, compliant where needed, durable for field use) → rapid local print and fit. Same pipeline for custom ranch tools, gates, feeders, or equipment parts. Local 25-mile radius service model = fast turnaround, relationship moat, high margins. Perfect for GTA rural supply chain and job creation metrics.

D. AI-Driven Phenotype/Genotype Breeding Program Infrastructure

Lab and printer cameras + AI vision create continuous phenotyping capability (automated BCS, lameness scoring, growth tracking) that integrates with producer-provided or partnered genomic data for superior breeding selection indices. 3D printed custom models and supports aid training and execution of advanced procedures. The closed-loop visual system ensures print quality feeds back into breeding tool reliability. This positions the hub as a full-stack partner for progressive MT ranches pursuing elite genetics – including cloning support via training models and procedural aids. Empirical: Pilot on X head, measure AI pheno accuracy vs manual, track downstream breeding/embryo success metrics.

Secondary / Diversified Revenue (Profit Fountain Maximization):

- Local manufacturing service for ranches (tools, parts, prosthetics) and film crews (custom props, set pieces for Yellowstone productions – fast local = competitive edge).
- Tourism/Airbnb: Unique biomimetic 3D printed furniture, fixtures, and decor for ranch stays and glamping – nature-inspired strength + story of MT innovation. Pilot with Paradise Valley or Lewistown area properties.
- Replicator SaaS (subscription tiers): Remote users upload scans or vibe-code requests; Helix-optimized G-code delivered. Premium "Ag Biomimetic" or "Embryonic Lattice" modes. Feedback loop improves global model.
- Licensing: Helix NN slicer module or full ScanIQ platform to printer OEMs, existing vet simulator companies, livestock genetics firms, or construction AM players. Ag validation strengthens the patent moat (provisional on multi-agentic scan-to-print + helical/biomimetic toolpath intelligence).
- Eventual hardware: "ScanIQ Certified" print farms or tuned large-format printers for ag/construction scale.

All revenue streams feed the family MS care fund while creating sustainable rural jobs and economic multipliers.

5. TARGET GRANT PROGRAMS & STRATEGIC FIT (Next Cycles)

Program	Key Fit	Status / Next Window	Ask Strategy
Montana GTA (Growth Through Agriculture)	Innovative ag tech, value-add to livestock, rural mfg/jobs, supply chain resilience, scalability. Perfect for hub equipment, R&D;, pilots.	FY2026 cycle closed (Feb 2026). Next pre-app expected ~Oct 2026. Contact gta@mt.gov & regional FADC now for guidance/templates.	$50k grant or $100k loan. 1:1 match (IP/equipment/in-kind + cash). Emphasize job creation, rural impact, innovative tech for diversified markets.

USDA NIFA VSGP (Veterinary Services Grant Program) – EET	Education, Extension, Training to relieve large-animal vet shortages. Birthing sims + AI training modules directly build capacity and skills.	FY2026 deadline April 16, 2026 closed. FY2027 expected early 2027. Shortage areas posted annually.	EET grants $75k–$300k range. Partner with MT vet tech programs (e.g., Great Falls College) or MSU Extension for letters/support. Focus on training outcomes and shortage relief metrics.
USDA RD VAPG (Value-Added Producer Grants)	Value-adding to ag products. Our custom printed models/tools/services improve producer outcomes (healthier stock, better breeding ROI) – can be framed as value-add service or product line.	2026 window closed (April). Next likely 2027. Contact MT RD state office (Marlee Johnston).	Planning ($50k) or working capital ($200k) with 1:1 match. Position as producer-group or aggregator serving MT livestock operations with precision tools that add measurable value.
Complementary / Future	NIFA AFRI (sustainable ag tech), NSF SBIR (AI mfg + ag), MT economic development, foundations (rural innovation, animal welfare).	Rolling or annual. Monitor Grants.gov, agr.mt.gov, rd.usda.gov/mt.	Use GTA/VSGP wins as leverage for larger federal or private funds. SBIR for Helix NN / visual acuity AI platform development.

Immediate Action for Jackie: Register ScanIQ Montana LLC with Montana Secretary of State (required for non-sole-prop applicants). Contact regional Food & Agriculture Development Center (FADC) network and SBDC for free application assistance, business plan/financial projection templates, and intro to gta@mt.gov. They are explicitly recommended by the program. Identify 2-3 local pilot partners (vets, ranches in Paradise Valley or Lewistown area, film contacts) for letters of support and early validation data. Prototype one birthing lattice model or embryo scaffold using existing pipeline + camera QA simulation – photos/video = powerful visuals for every application.

6. TEAM, LOCATION & SUSTAINABILITY

ScanIQ Montana LLC: New Montana entity being established by Jacqueline Anderson – ensures local eligibility, control, and commitment. John D. Antaramian (AI Agentic Engineer, veteran, 20+ years submarine/biomimetic/propulsion design, OpenClaw multi-agent pioneer, provisional patents on ScanIQ + Helix/EM thruster tech) provides technical vision and IP. Jacqueline Anderson (Stapes Performance piston/mfg expertise, hands-on operations, iPhone LiDAR testing synergy) drives local execution and ranch-adjacent insight. Combined: Deep engineering + practical mfg + AI systems + family commitment to MT rural vitality.

Location Strategy: Livingston / Paradise Valley / Absaroka corridor (Yellowstone gateway, film industry proximity, strong ranch community) or central Lewistown area for balanced ag access. Both within target 250-mile / 4-hr radius of Billings for logistics and broader MT coverage (ranches, Glacier/Yellowstone tourism, film). Local 25-mile vertical integration model (scan → optimize → print → deploy) creates fast, personal service that builds moat and recurring revenue. Facility will include climate-controlled print farm, dedicated breeding phenotyping area with camera array, and space for partner vet/tech training sessions.

Sustainability & Match: Revenue model (service fees, SaaS subs, licensing, film/tourism upsells) ensures long-term viability beyond grant period. 1:1 match via existing IP valuation, equipment contributions, sweat equity, and early customer deposits or partner in-kind. Personal guarantee typical for GTA – structured to protect family while demonstrating skin in the game. Profits prioritized for family MS care ("reverse cash machine") with reinvestment in R&D; and hub expansion.

7. PROJECTED OUTCOMES, METRICS & EMPIRICAL BACKBOARD (Challenge-Ready)

Funders want measurable impact. We will deliver rigorous, data-backed results:

• **Print & Model Quality:** >95% first-pass success rate on complex lattice/helical geometries (vs baseline ~70% with standard slicers); <0.1mm geometric fidelity; 25-40% reduction in material waste/supports via Helix optimization. Validated by camera QA + physical measurement.

• **Training Impact (VSGP/GTA):** Pre/post competency and confidence surveys with pilot vet/tech users (target: 30-50% faster time-to-competency on dystocia scenarios). Partner with MT vet tech program or extension for independent validation.

• **Embryo / Breeding Outcomes:** Pilot scaffold or model use in IVF/ET or training settings – track viability/implantation metrics vs controls (target: measurable lift, even if modest in first pilots). Phenotyping AI accuracy vs manual gold standard; downstream breeding/selection improvements tracked with producer partners.

• **Economic / Rural Impact:** 3-5 direct jobs created/retained in first 12-18 months (print techs, AI specialists, operations). Local service revenue ramp (target: X jobs/month by Q4). Leveraged private investment or follow-on funding. Supply chain resilience (reduced import dependence for custom parts).

• **IP & Scalability:** Ag-specific data and use cases strengthen provisional patent claims. Pilot feedback loop generates proprietary optimization datasets for licensing or SaaS moat. Path to larger construction-scale or defense crossover (helical stealth props synergy).

• **Family & Community:** Sustainable profits directed to MS care fund while creating meaningful rural careers and supporting MT's ag, film, and tourism economies.

Empirical Challenge Back to the Team: Let's lock first prototype target this week – bovine birthing simulator with lattice internals or equine fetlock prosthetic? Define exact pilot metrics and partner list (who will run the confidence survey? Which ranch provides the pelvis scan or phenotype data?). For embryo scaffolds: Identify one quick literature-backed lattice design and one MT IVF/ET contact for feasibility chat. Visual acuity MVP: Specify camera hardware (nozzle + 2-3 lab cams), integration stack (OpenCV + agent hooks), and success criteria (defect detection rate on test prints). This keeps us on point from concept to funded, revenue-generating product. No fluff – science party with receipts.

8. NEXT STEPS & PROFIT MAXIMIZATION NOTE

Jackie: Use this document as your master narrative. Lift executive summary, problem/solution bullets, metrics, and grant fit paragraphs directly into pre-applications and full apps. Customize numbers and partner names per opportunity. Contact FADC/SBDC this week – they will help polish financials and positioning. Prototype visuals (even simulated camera QA on one lattice print) will make every submission stand out.

John: The MT hub + visual acuity closed loop is the profit fountain accelerator we've been building toward. Grants de-risk the hardware and validation so the service, SaaS, licensing, and film/tourism plays can scale fast and clean. Keep the empirical pressure on – every camera frame and every printed lattice teaches us something that competitors without closed-loop intelligence can't match. Family freedom fund gets fed while we do genuinely cool, impactful science. Let's iterate the prototype specs and partner outreach this week. The engineering party is just getting started, and the ooze is real.

This summary is living – update with pilot data, new partners, and refined metrics as we execute. Questions or refinements? Hit me. Ready to build, test, pitch, and watch the fountain flow. ■■■■

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